

LESSON 2-6 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2

Median and Outliers

Maryland Standard 6.SP.B.5d • Learning target – I can explain why the median is a better measure of center than the mean when a data set has an outlier.

REMEMBER

- An **outlier** is a value much greater or much less than the rest of the data.
- An outlier pulls the **mean** toward it. The **median** barely moves, so it is **resistant** to outliers.
- With an outlier, the median usually describes a **typical value** better. Still report the outlier; it is real data.

WORKED EXAMPLE

Six bike rides lasted 14, 18, 15, 16, 64, and 17 minutes. The mean is 24 minutes. Which measure better describes a typical ride?

Step 1 Order: 14, 15, 16, 17, 18, 64. The outlier is 64.

Step 2 Median: $(16 + 17) \div 2 = 16.5$. The mean, 24, is greater than 5 of the 6 rides.

Answer The median, 16.5 minutes

WORD BANK

outlier – a number far away from the others

median – the middle number

mean – add all, then divide by how many

typical – usual, like most of the data

PRACTICE

show your work • label units

1. Find the median of each data set. Circle the outlier in each set.

Step 1 Order each list.

Step 2 Circle the number far from the rest.

Step 3 Find the middle number.

a. 9, 11, 10, 12, 13, 8, 48 _____

b. 3, 70, 72, 75, 78, 71, 74 _____

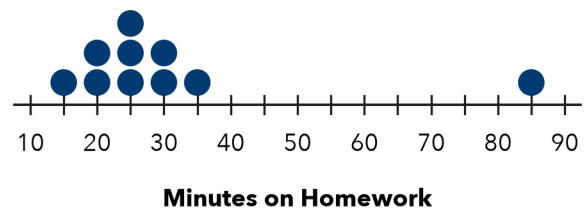
c. 25, 28, 30, 31, 33, 95 _____

d. 40, 41, 43, 44, 46, 47, 150 _____

e. 6, 7, 7, 8, 9, 10, 34 _____

f. 56, 52, 55, 54, 9, 53, 57, 58 _____

- 2. Homework Minutes.** The dot plot shows minutes 10 students spent on homework one night. The mean is 31 minutes.



Step 1 Find the dot far from the others.

Step 2 Order the 10 values; the median is between the 5th and 6th.

Step 3 Count students below 31.

- a. Outlier: _____
- b. Median: _____ min
- c. Students below the mean: _____
- d. Better center: _____

- 3. Library Visitors.** Visitors on 7 days: 64, 58, 61, 66, 59, 63, and 4 (the library closed early for snow).

- a. Median of all 7 days: _____
- b. Median without the 4: _____

- 4. Car Wash.** A club earned \$10, \$12, \$13, \$14, \$15, \$16, and \$60 on 7 Saturdays.

Step 1 Mean: add the 7 amounts, then divide by 7.

Step 2 Median: the 4th amount in order.

- a. Mean: \$ _____
- b. Median: \$ _____
- c. Better center: _____

- 5. Find the Error.** Andre's data are 31, 33, 34, 36, 37, 39, and 210. He says, "The outlier 210 makes the median much bigger, so the median is not a good measure." Is he right? Find the median to support your answer.

Frame Andre is _____. The median is _____. Without 210 it would be _____.

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME Ask someone at home about a normal trip and their slowest trip. Which trip is an outlier?

LESSON 2-6 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Comparing Distributions**

Maryland Standard 6.SP.B.5d • Learning target – I can compare two distributions using the median and the interquartile range together.

REMEMBER

- Compare **centers** with the medians. Compare **spreads** with the IQRs ($IQR = Q3 - Q1$).
- A smaller IQR means more **consistent** data. A full comparison names **both** measures for **both** groups, with numbers.

WORKED EXAMPLE

Team Red: median 14 points, IQR 6. Team Blue: median 18 points, IQR 15. Compare the teams.

Step 1 Centers: $18 > 14$, so Blue usually scores more.

Step 2 Spreads: $6 < 15$, so Red is more consistent.

Answer **Blue scores more; Red is more consistent.**

WORD BANK

median – the typical value

IQR – $Q3 - Q1$, the spread of the middle half

consistent – close together

predictable – you can guess it well

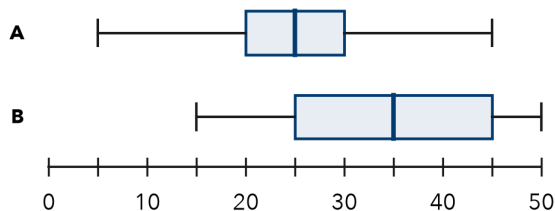
PRACTICE

show your work • label units

- 1. Reading Minutes.** The box plots show minutes read per day by Class A and Class B.

Step 1 Find the line in each box: the medians.

Step 2 $IQR = \text{right edge} - \text{left edge}$ of each box.



- Medians: A ____ B ____
 - IQRs: A ____ B ____
 - Usually reads more: _____
 - More consistent: _____
- 2. Bus Routes.** Minutes late on 7 days. **Route 3:** 2, 3, 5, 6, 8, 9, 11. **Route 8:** 0, 1, 4, 6, 10, 14, 15.
- Step 1** Order is given. The median is the 4th value.
- Step 2** $Q1$ is the 2nd value, $Q3$ is the 6th value.
- Route 3: median ____ IQR ____ Route 8: median ____ IQR ____
 - Which route is more predictable? Why? _____
- 3. Plant Heights (cm).** **Sun:** 10, 12, 13, 15, 16, 18, 19, 21. **Shade:** 6, 7, 8, 8, 9, 10, 10, 12.
- Sun: median ____ cm IQR ____ cm
 - Shade: median ____ cm IQR ____ cm

4. **Delivery.** Shop K: median 28 min, IQR 8 min. Shop L: median 36 min, IQR 8 min. What is the same?

☐ the same center ☐ the same spread ☐ both center and spread

5. **Find the Error.** Set E: median 70, IQR 25. Set F: median 64, IQR 6. Jada says, "Set E is higher and more consistent." Explain her error.

Frame Set E has the higher _____. The more consistent set is _____ because its _____ is _____.

TEST PRACTICE

choose the best answer

6. Group 1: median 12 cm, IQR 3 cm. Group 2: median 12 cm, IQR 9 cm. Which is more consistent?

A Group 1 **B** Group 2 **C** Both the same **D** Cannot tell

7. What is the IQR of 3, 5, 6, 8, 10, 12, 13?

A 7 **B** 8 **C** 10 **D** 17

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME Tell someone at home why a bus with a small IQR is easier to plan around.